Politecnico di Torino

Master of Science (M.Sc.) in Computer Engineering

Academic Year 2025/2026

Graduation Session March 2026

Design and Implementation of a Cloud-Native Reactive Architecture

The Smart Travel Platform Case Study

Supervisor

Prof. Giovanni Malnati

Candidate

Federico Palazzi

Abstract

The rapid evolution of enterprise web applications has driven a paradigm shift from monolithic architectures to cloud-native microservices solutions. This transition is motivated by the need for greater scalability, resilience, and agility in the software development life cycle. This thesis presents the design, development and deployment of "Smart Travel", a comprehensive e-commerce platform dedicated to the sale of travel packages. The project serves as a proof of concept developed within an industrial internship context, aiming to analyze the complexities and benefits of a modern and distributed software architecture.

The primary objective of this work is not merely the implementation of functional requirements (such as catalog management and order processing), but rather the critical evaluation of competing technologies and the automation of the release process. A central element of this research is the comparative analysis between two leading Java ecosystems: the established Spring Boot framework and the cloud-native Quarkus framework. Specifically, the project adopts a fully reactive approach for both of them, utilizing Spring WebFlux and Quarkus Reactive capabilities.

From an architectural perspective, the "Smart Travel" system is designed to ensure loose coupling and high availability. The backend adopts a microservices pattern supported by RabbitMQ, a message broker that facilitates asynchronous event-driven communication among services. Data persistence is managed through MongoDB, a NoSQL database chosen for its schema flexibility, which is particularly advantageous for handling diverse and evolving travel package structures compared to rigid relational models. On the client side, the user interface is delivered via a Single Page Application (SPA) developed with Angular and leveraging RxJS for reactive state management, ensuring a responsive and fluid user experience.

A significant portion of this thesis is dedicated to the "industrialization" of the software, bridging the gap between development and operations (DevOps). The project moves beyond local execution by implementing a robust Continuous Integration and Continuous Delivery (CI/CD) pipeline using Jenkins. This pipeline automates the entire delivery chain, including code checkout, compilation, container image creation and pushing to a private registry. The final deployment target is OKD (the community distribution of Kubernetes that powers Red Hat OpenShift). The thesis details the orchestration of containerized applications within the OKD cluster, managing critical aspects such as service discovery, route exposure and configuration management via ConfigMaps and Secrets.

In conclusion, the "Smart Travel" project demonstrates the viability and robustness of a heterogeneous, fully reactive microservices architecture. It successfully integrates different modern technologies into a cohesive ecosystem, proving that the

adoption of cloud-native principles, while introducing complexity in orchestration and consistency management, offers superior long-term benefits in terms of maintainability and scalability. The work provides a practical blueprint for modernizing legacy Java applications, offering concrete evidence of how the synergy between Reactive frameworks, Event-Driven patterns, and automated DevOps practices defines the state of the art in contemporary software engineering.

Acknowledgements

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Tristique senectus et netus et malesuada fames ac turpis. Pretium nibh ipsum consequat nisl vel pretium lectus quam. Urna molestie at elementum eu facilisis sed odio morbi quis. Sed egestas egestas fringilla phasellus faucibus scelerisque eleifend. At in tellus integer feugiat. Mauris rhoncus aenean vel elit scelerisque mauris pellentesque pulvinar. Commodo sed egestas egestas fringilla. Nunc lobortis mattis aliquam faucibus purus in massa. Facilisis magna etiam tempor orci eu lobortis elementum nibh. Elementum curabitur vitae nunc sed velit dignissim. Neque volutpat ac tincidunt vitae. Massa id neque aliquam vestibulum morbi blandit cursus risus at. Porta non pulvinar neque laoreet suspendisse interdum consectetur. Turpis in eu mi bibendum. Ut tristique et egestas quis ipsum suspendisse. Integer quis auctor elit sed vulputate mi sit amet. Viverra nam libero justo laoreet sit amet cursus.

Table of Contents

List of Figures V

Glossary VII

1 Introduction 1
1.1 Goal 1
1.2 Thesis structure 2
1.2.1 Subsection 2

Bibliography 3

List of Figures

Glossary

DBMS

Database Management System

Chapter 1

Introduction

This is an example citation [1].

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Porttitor eget dolor morbi non arcu risus quis varius. Libero id faucibus nisl tincidunt. Neque laoreet suspendisse interdum consectetur libero id faucibus nisl tincidunt. Scelerisque in dictum non consectetur a erat. Leo a diam sollicitudin tempor id eu. Sodales ut eu sem integer vitae justo eget magna fermentum. A cras semper auctor neque vitae. Cursus euismod quis viverra nibh cras pulvinar. Mi tempus imperdiet nulla malesuada pellentesque elit eget gravida cum. Dictum at tempor commodo ullamcorper a lacus vestibulum sed. Ultricies mi eget mauris pharetra. In pellentesque massa placerat duis ultricies lacus. Fringilla phasellus faucibus scelerisque eleifend donec pretium vulputate.

Aliquam sem fringilla ut morbi tincidunt augue interdum. Risus at ultrices mi tempus imperdiet nulla malesuada pellentesque. Semper quis lectus nulla at volutpat. Nullam non nisi est sit amet facilisis. Eget velit aliquet sagittis id consectetur purus ut faucibus pulvinar. Ultricies tristique nulla aliquet enim tortor at auctor urna nunc. Pharetra diam sit amet nisl suscipit adipiscing bibendum est ultricies. Amet facilisis magna etiam tempor. Nunc non blandit massa enim nec dui nunc mattis. At ultrices mi tempus imperdiet nulla malesuada pellentesque. Cursus metus aliquam eleifend mi in nulla posuere. In eu mi bibendum neque egestas congue quisque. Augue eget arcu dictum varius duis at consectetur.

1.1 Goal

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Tristique senectus et netus et malesuada fames ac turpis. Pretium nibh ipsum consequat nisl vel pretium lectus quam. Urna

molestie at elementum eu facilisis sed odio morbi quis. Sed egestas egestas fringilla phasellus faucibus scelerisque eleifend. At in tellus integer feugiat. Mauris rhoncus aenean vel elit scelerisque mauris pellentesque pulvinar. Commodo sed egestas egestas fringilla. Nunc lobortis mattis aliquam faucibus purus in massa. Facilisis magna etiam tempor orci eu lobortis elementum nibh. Elementum curabitur vitae nunc sed velit dignissim. Neque volutpat ac tincidunt vitae. Massa id neque aliquam vestibulum morbi blandit cursus risus at. Porta non pulvinar neque laoreet suspendisse interdum consectetur. Turpis in eu mi bibendum. Ut tristique et egestas quis ipsum suspendisse. Integer quis auctor elit sed vulputate mi sit amet. Viverra nam libero justo laoreet sit amet cursus.

1.2 Thesis structure

Lorem ipsum dolor sit amet:

- onsectetur adipiscing elit,
- sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
- Tristique senectus et netus et malesuada fames ac turpis.

1.2.1 Subsection

Sottosezione

Bibliography

[1] European Union Agency for Cybersecurity (ENISA). *ENISA Threat Landscape: Finance Sector*. 2025. URL: https://www.enisa.europa.eu/sites/default/files/2025-02/Finance%20TL%202024_Final.pdf (cit. on p. 1).